Introduction to Cloud Security-Module 1

Introduction-Cloud Computing

- Cloud Computing is the delivery of computing services (servers, storage, databases, networking, software) over the Internet on a pay-as-you-go basis.
- Key terms:
  - Internet-based
  - On-demand
  - Scalable
  - Pay-per-use
- Cloud Services
  - Google Drive(Saas)
  - Gmail
  - Amazon Web Services (Infrastructure, platform…)
  - Microsoft Azure
  - Netflix
  - Instagram
  - Youtube

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Traditional Computing vs Cloud Computing

Traditional (On-Premise)

- You buy your own server
- Install software
- Maintain hardware
- Pay electricity & maintenance
- Hire IT staff

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Example: College maintaining its own exam server.

Cloud Computing

- Rent servers from cloud provider
- Access via Internet
- Pay only for usage
- No hardware maintenance

Example: Using Google Drive instead of pen drive.

Example 1

- When you upload a photo on Instagram:

Your phone sends the photo

Through the Internet

To Instagram's servers

- Your data does NOT stay in your phone.

It Travels :Your Phone → Internet → Instagram Server (Cloud Data Center)

1. **Cloud Data Center** means: A large physical building that contains thousands of powerful computers (servers) connected to the internet, used to store and process data for cloud services.
2. Instagram does not store your photo on a single computer.
3. Instead, it stores it in: Massive data centers, Thousands of servers, Located in different countries, Managed by cloud infrastructure

Example 1(cont'd)

- What Happens Inside the Cloud?
- When you upload a photo:
- **The image is saved on multiple servers**
- (For backup and reliability)
- **It is stored in databases**
- (So it can be searched and retrieved)
- **It is processed**
- (Compression, filters, tagging- attaching keywords to search, AI detection-AI to auto identify and label content)
- **It is distributed globally**
- So your friend in another country can see it instantly.

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When a user posts on Instagram, the data is transmitted via the Internet and stored in distributed cloud data centers, where it is replicated, processed, and made globally accessible.

Why Use Cloud Data Centers?

- Because:

Millions of users upload photos every second

Data must be available 24/7

System must handle high traffic

Storage must scale automatically

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- If Instagram stored everything on one local computer:
  It would crash
  It would run out of storage
  It would not support global users
- These data centers are real physical buildings located in different parts of the world.
- “Cloud means data floating in the sky.”---wrong

Example 2: AWS for Infrastructure

Amazon EC2 allows a customer to rent a virtual server over the Internet.

Instead of buying a physical server, customer can:

- Choose CPU size(No of virtual Cpus)
- Choose RAM
- Choose storage
- Select Operating System (Linux/Windows)
- Launch the server
- Pay per hour/second

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That is Infrastructure as a Service (IaaS).

"Imagine the customer is building a college website."

Instead of buying: Physical server, UPS, Cooling system, Network setup

The customer simply:

Log into AWS, Launch an EC2 instance, Install Apache(Web Server sw / Node.js, so that his website can run and be accessible on the internet.

Cloud Computing

- **Cloud Computing** is a technology that allows a customer to store and access data and applications over the internet instead of using his computer's hard drive or a local server.
- **Cloud computing** is a service provisioning technique where computing resources like hardware such as servers and storage devices, software's and complete platform for developing applications are provided as a service by the cloud providers to the customers.
- **Customers** can use these resources as and when needed, can increase or decrease resource capacities dynamically according to their requirements and pay only for how much the resource were used.

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Cloud Service Provider (CSP)

- A **Cloud Service Provider (CSP)** is a company that provides cloud computing services like storage, servers, networking, databases, software, and more — over the internet. Think of a CSP as a **company that owns and manages data centers**, and the user rent their resources instead of buying their own hardware.

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- Some Examples:

- **Amazon Web Services (AWS): Popular for** Services like EC2, S3, RDS
- **Microsoft Azure:** widely used by enterprises, offering strong integration with Windows, Microsoft software, and business applications.
- **Google Cloud Platform:** known for advanced AI, machine learning, and large-scale data analytics solutions.
- **IBM Cloud: f**ocused on enterprise applications, hybrid cloud environments,
- **Oracle Cloud: Focused on** Database-focused cloud services

Cloud Computing Solves: The "Old Way" vs. The "New Way"

- Before the cloud, if a company/customer wanted to launch a website or an application, it had to follow a slow and expensive process:
- **The "Old Way" (On-Premises):** The company had to buy powerful physical servers, storage disks, and networking equipment. This required a huge upfront investment, known as **Capital Expenditure (CapEx)**. If they needed a new server, the process of ordering, installing, and configuring it could take weeks or even months.
- **The "New Way" (Cloud Computing):** The cloud changes this model entirely. Instead of buying hardware, you rent computing power from a cloud provider (like AWS, Google, or Microsoft). This shifts the cost from a large upfront investment to a manageable monthly bill, known as an **Operational Expenditure (OpEx)**. This model eliminates long waiting times, and wasted resources.

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I need to grow my infrastructure, but I do not know for how long…
I have a lot of infrastructure that I want to rent …
I have a surplus of infrastructure that I want to make use of
I cannot invest in infrastructure, I just started my business….
I have infrastructure and middleware and I can host applications
Global Cloud Marketplace
I want to focus on application logic and not maintenance and scalability issues
I have infrastructure and provide application services
I want to access and edit my documents and photos from everywhere..

The 5 Core Characteristics of Cloud Computing

All cloud services share five fundamental characteristics that define them:

- **On-Demand Self-Service:** Users can provision computing resources like servers and storage automatically, without requiring human intervention from the service provider.
- **Broad Network Access:** Cloud services are available over the internet and can be accessed from anywhere using different types of devices through standard network methods.
- **Resource Pooling:** The provider's computing resources are pooled to serve multiple customers using a multi-tenant model-shared by multiple customers. Resources are dynamically assigned and reassigned according to demand.
- **Rapid Elasticity (Scalability):** Resources can be scaled up or down quickly and, in some cases, automatically, to meet demand. This ensures enough computing power(CPU processing power, Memory (RAM), Storage, Network bandwidth) (to handle many users or requests) during traffic spikes and aren't paying for idle resources during quiet periods.
- **Measured Service (Pay-as-you-go Model):** Cloud systems automatically control and optimize resource use. Usage is monitored, controlled, and reported, providing transparency for both the provider and the consumer. Users only pay for what they use.

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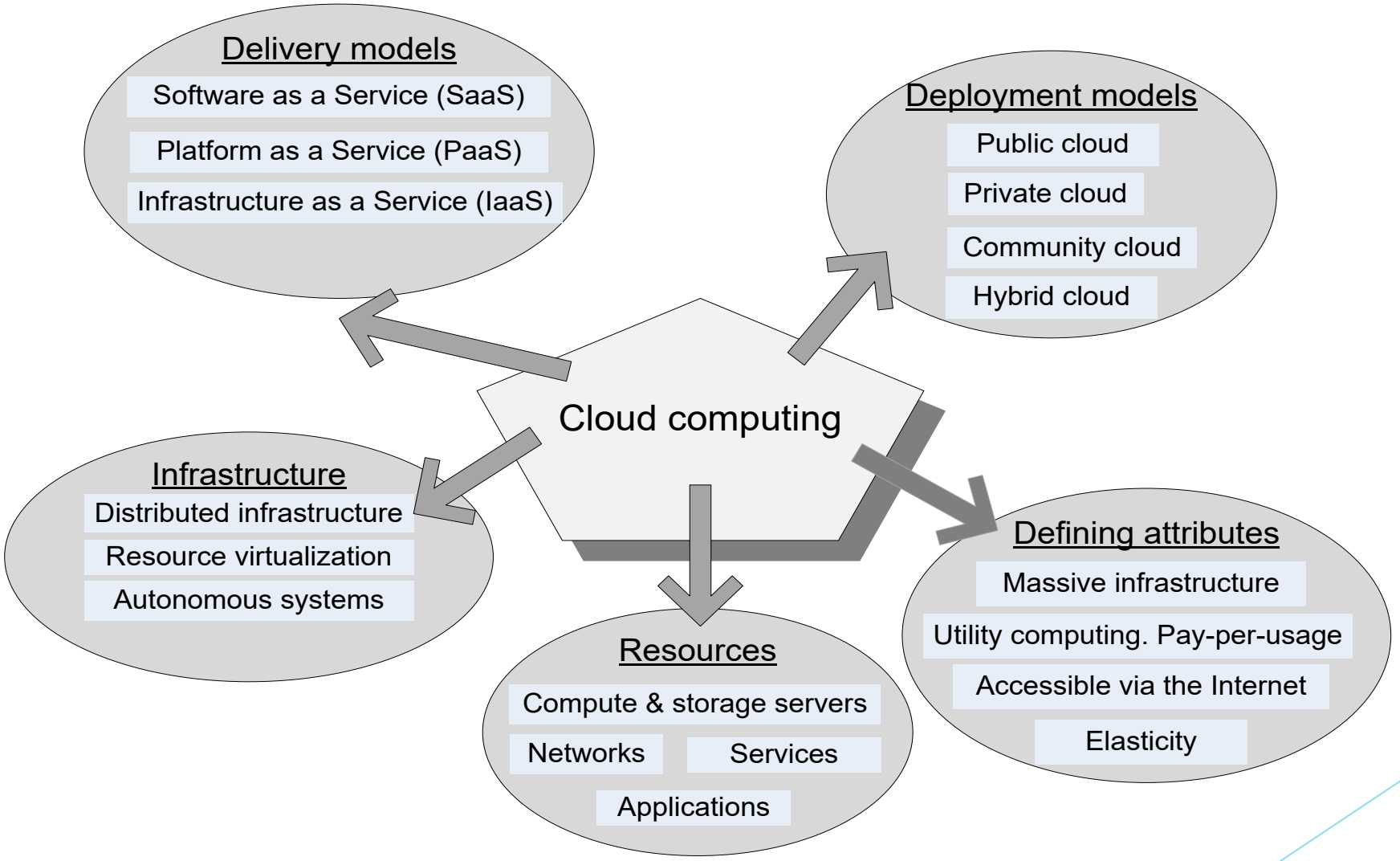
Delivery models
Software as a Service (SaaS)
Platform as a Service (PaaS)
Infrastructure as a Service (IaaS)
Deployment models
Public cloud
Private cloud
Community cloud
Hybrid cloud
Cloud computing
Infrastructure
Distributed infrastructure
Resource virtualization
Autonomous systems
Resources
Compute & storage servers
Networks
Services
Applications
Defining attributes
Massive infrastructure
Utility computing. Pay-per-usage
Accessible via the Internet
Elasticity

Cloud delivery models

- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)

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Cloud delivery models: IaaS

Infrastructure as a Service (IaaS) provides virtualized computing resources such as servers, storage, and networking over the internet. The user manages the operating system and applications, while the cloud provider manages the hardware.

- What the User Manages:

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Operating System
Applications
Data

- What the Provider Manages:

Physical servers
Networking
Storage hardware

- Popular Examples: Amazon Web Services – EC2, Microsoft Azure – Virtual Machines

Cloud delivery models: PaaS

Platform as a Service (PaaS) provides a ready-to-use platform with development tools, runtime environment, and deployment support so developers can build and run applications without managing servers.

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- What the User Manages:

Application code

Data

- What the Provider Manages:

Servers

Operating system

Runtime

Popular examples of runtime environment: JVM (Java Virtual Machine), Node.js runtime, Python interpreter

Cloud delivery models: SaaS

Software as a Service (SaaS) provides ready-to-use software applications over the internet. Users simply access the software through a web browser without installing or managing it.

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- What the User Manages:

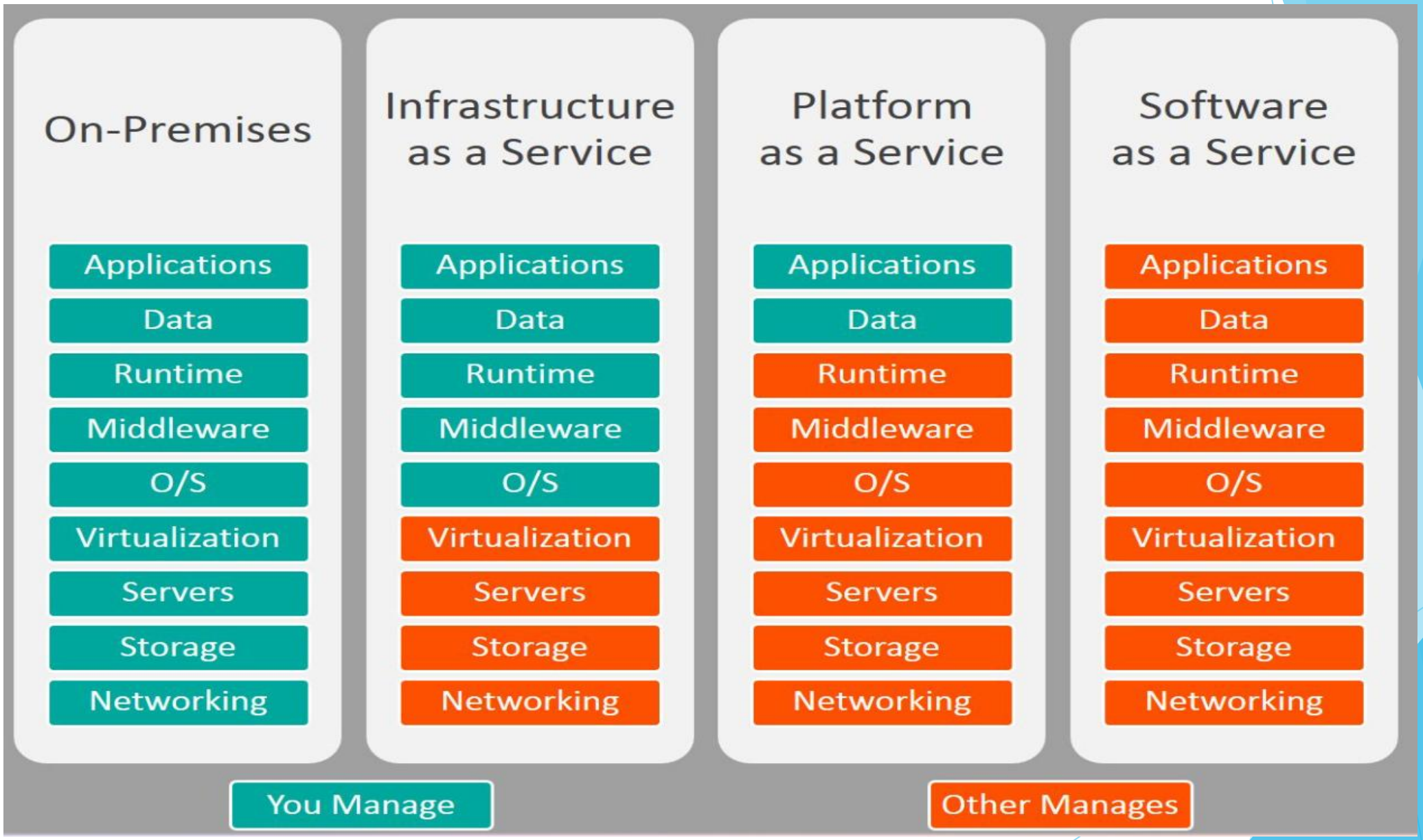
Only usage (no technical management)

- What the Provider Manages:

Application
Servers
Database
Security
Updates

Popular Examples:

Google Workspace – Google Docs / Gmail


On-Premises
Infrastructure as a Service
Platform as a Service
Software as a Service
Applications
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
Applications
Data
Runtime
Middleware
O/S
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Servers
Storage
Networking
Applications
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
Applications
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
You Manage
Other Manages

Cloud Deployment Model

- Whether you're a small business or a large enterprise, choosing the right cloud model can affect your security, scalability, and operational efficiency.
- A cloud deployment model fundamentally determines where the infrastructure for your deployment exists and who owns and controls that infrastructure.
- The location of the servers you're utilizing and who controls them are defined by a cloud deployment model. It specifies how your cloud infrastructure will look, what you can change, and whether you will be given services or will have to create everything yourself.

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Cloud Deployment Model-Public Cloud

- The public cloud makes it possible for anybody to access systems and services. The public cloud may be less secure as it is open to everyone.
- The public cloud is one in which cloud infrastructure services are provided over the internet to the general people or major industry groups.

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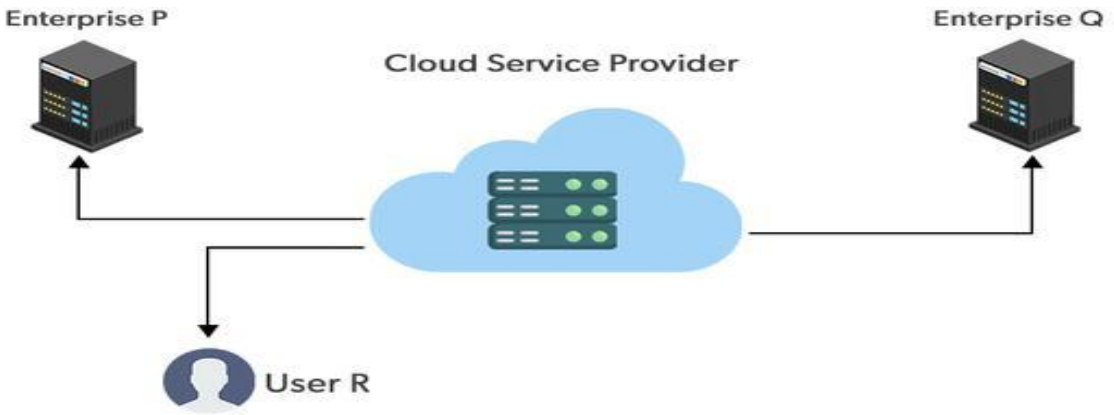
- The infrastructure in this cloud model is owned by the entity that delivers the cloud services, not by the consumer. It is a type of cloud hosting that allows customers and users to easily access systems and services.

Gmail and Google App Engine are examples of Public Cloud.

- Gmail is a public cloud service because it is owned by Google and available to anyone over the Internet.
- Google App Engine is a public cloud platform where developers can deploy applications on Google's shared cloud infrastructure.

Public Cloud



Cloud Deployment Model-Private Cloud

- The private cloud allows only a single organization to access systems and services. The private cloud is more secure as it is restricted to a specific organization.
- The private cloud is one in which cloud infrastructure services are provided to a single organization, whether managed internally or by a third party.

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- The infrastructure in this cloud model is owned or exclusively used by one organization, not shared with the general public. It is a type of cloud hosting that provides greater control and security.

Examples of Private Cloud: VMware Private Cloud and OpenStack Private Cloud

Large enterprises like banks, telecom companies, and manufacturing firms run internal applications on VMware-based private clouds inside their own data centers.

Government agencies and research institutions use OpenStack to build their own internal cloud infrastructure.

Large banks like **JPMorgan Chase** build internal private cloud infrastructure to run:

- Core banking systems
- Payment processing
- Risk analysis systems

These systems cannot be hosted on public cloud due to regulatory and security reasons.

Cloud Deployment Model-Hybrid Cloud

- The hybrid cloud is a combination of public and private cloud. It allows data and applications to be shared between them.
- The hybrid cloud is one in which some services are hosted on a private cloud while others are hosted on a public cloud.
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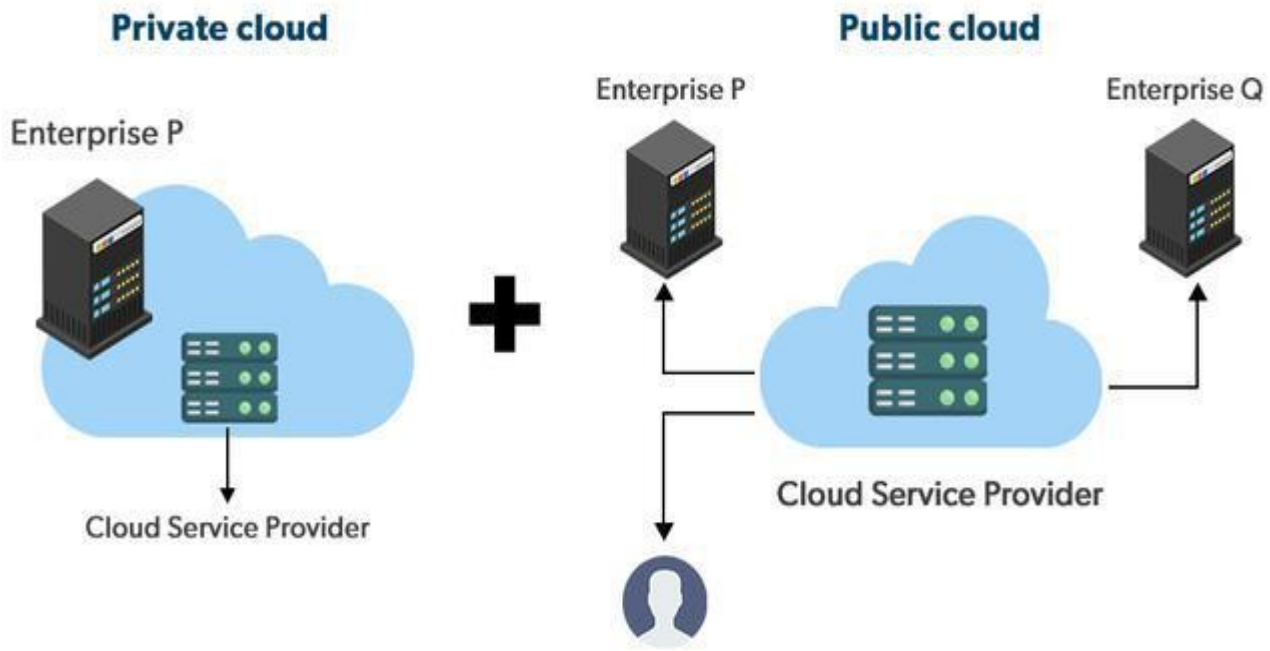
- The infrastructure in this cloud model combines both privately owned resources and third-party cloud services. It provides flexibility and better scalability.
- Netflix is a classic example often discussed in cloud studies.
  - Runs most of its streaming infrastructure on Amazon Web Services (Public Cloud)
  - Maintains internal systems for content production, billing, and corporate operations (Private Infrastructure)
  - Public cloud for scalability
    Internal systems for control
- Combination = Hybrid Cloud


Private cloud
Enterprise P
Cloud Service Provider
Public cloud
Enterprise P
Enterprise Q
Cloud Service Provider

Hybrid Cloud
Private Cloud
Public Cloud

Cloud Deployment Model-Community Cloud

- The community cloud allows access to systems and services by a group of organizations with shared concerns.
- The community cloud is one in which cloud infrastructure services are shared by several organizations that have common objectives, security requirements, or policies.
- The infrastructure in this cloud model is shared among specific organizations, not open to the public. It is suitable for collaboration and regulatory compliance.
- A government cloud shared by different government departments is an example of Community Cloud.
- A cloud infrastructure shared by multiple hospitals for medical data exchange is also an example of Community Cloud.

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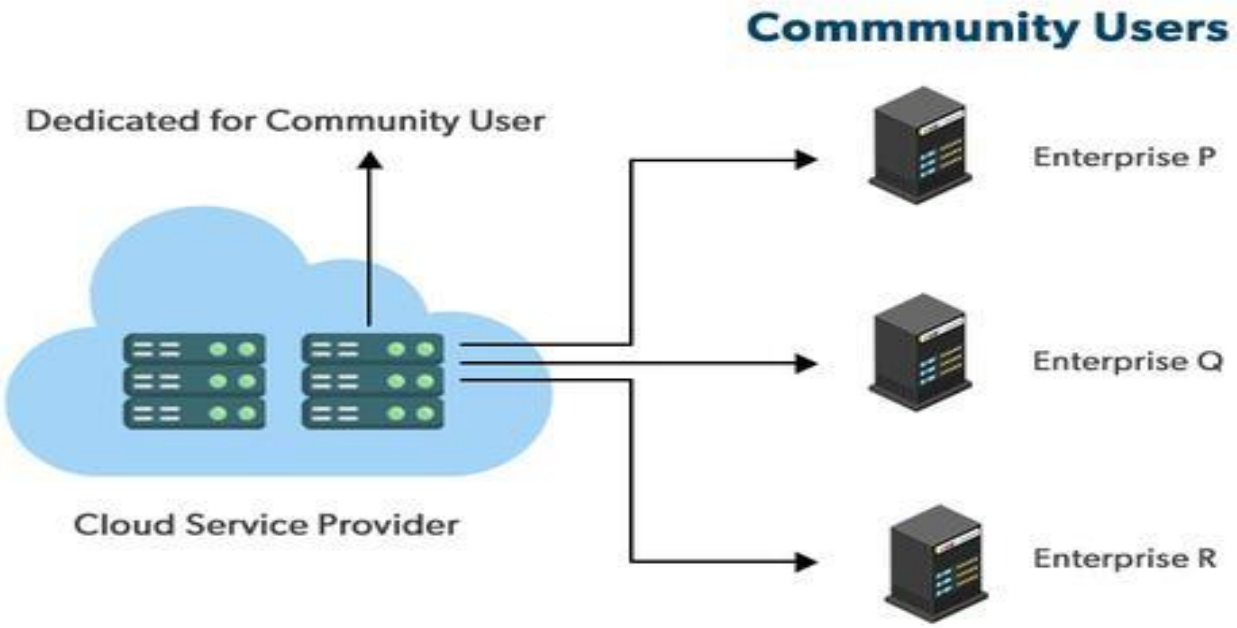
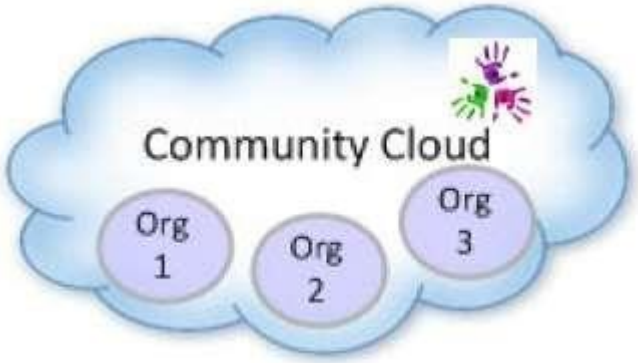

Commmunity Users
Dedicated for Community User
Enterprise P
Enterprise Q
Enterprise R
Cloud Service Provider


Community Cloud
Org 1
Org 2
Org 3

Deployment Models-Summary

- Public Cloud - the infrastructure is made available to the general public or a large industry group and is owned by the organization selling cloud services.

- Private Cloud – the infrastructure is operated solely for an organization.

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- Hybrid Cloud - composition of two or more clouds (public, private, or community) as unique entities but bound by standardized technology that enables data and application portability.

- Community Cloud - the infrastructure is shared by several organizations and supports a community that has shared concerns.

Shared Responsibility Model

Security in the cloud is a partnership between the cloud provider and the customer. This is known as the Shared Responsibility Model.

•**The Cloud Provider (e.g., AWS, Azure, GCP)** is responsible for the security *OF* the cloud. This includes the physical security of the data centers, the hardware, and the core networking infrastructure.

•**The Customer(user)** is responsible for security *IN* the cloud. This includes managing who has access to the resources, encrypting data, configuring firewalls (like Network Policies), and securing the applications of customer.

Simple Example

Imagine when a web app is deployed on AWS EC2.

AWS is responsible for :

Server, Network, Storage

User is Responsible for :

- **Application Configuration**: Application Configuration refers to the settings that are configured by the customer while building and using their application.
- These include database settings, login rules, access permissions, HTTPS configuration, file upload limits, etc

- **Security Responsibilities** : Security Responsibilities of the Customer refer to the obligation of the customer to securely manage and use cloud resources.

- These include Managing user access (IAM users & roles), Enforcing strong passwords and MFA, Enabling encryption, Configuring firewalls, Securing application code, Preventing misconfigurations

Security OF the Cloud.

The cloud provider is responsible for protecting the **infrastructure that runs the cloud**.

They manage:

Physical data centers
Servers and hardware
Storage devices
Networking cables & routers
Power supply & cooling
Hypervisor / virtualization layer

If AWS has a data center in Mumbai:

They secure the building
They install CCTV cameras
They control physical entry
They protect servers from fire or flood
Users do NOT manage this.

That is **Security OF the Cloud**

Security IN the Cloud: Application Configuration

1. Configure Database Username & Password: Users need to create and manage secure credentials for databases they provision. User must create strong database credentials. Change default passwords. Do not use simple passwords like admin123.

2. Set File Upload Size Limits (Example: AWS S3): Define rules for file uploads. Example: In Amazon Web Services (AWS) S3, the user can set `"content-length-range"` policy. This prevents users from uploading very large or harmful files.

3. Configure Login Rules (Use MFA): Enable Multi-Factor Authentication (MFA). Users must provide: Password + OTP (SMS code), OR Password + Fingerprint. This makes login more secure.

4. Decide Who Can Access Admin Panel: Use Role-Based Access Control (RBAC). Example in AWS IAM: Admin → Full control. Editor → Create/Edit/View Viewer → Only View. Not everyone should get full access.

5. Enable HTTPS : To enable secure communication for user'application. Data is encrypted when it travels between browser to Web Server(in HTTP Browser → Server, Data travels as plain text- so hackers can hack it)

Security IN the Cloud:Security Responsibilities

Encrypting Data: Cloud provider gives encryption tools /options to enable encryption for:

- Storage
- Databases

•The tools need to be enabled else data may be exposed.

Configuring Firewalls: To control incoming and outgoing traffic.

•In AWS, use Security Groups for Example:

- Allow HTTP (port 80)
- Block unknown ports

Securing Applications: Application code must be secure.

Example mistake: Login form without input validation. Hackers can inject malicious code

Preventing Misconfigurations: Properly set up and manage cloud resources so that they do not accidentally become insecure or publicly exposed.

Example: When sensitive data is in cloud storage. If it is accidentally made public. Data leaks.

Virtualization

Virtualization is the technology that allows one physical computer to run multiple virtual computers.

Instead of:

1 Server → 1 Operating System

We now have:

1 Server → Multiple Virtual Machines (VMs)

Simple Example
Imagine:

- One powerful physical server
- Divided into 5 smaller virtual servers
- Each virtual server runs its own OS
- This is Virtualization.

Basic Risks in Cloud (Multi-Tenancy)

- Multi-tenancy means **multiple users (tenants) share the same cloud infrastructure.**
- Many customers share the same:
  - Physical servers
  - Storage
  - Network resources

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- Isolation is maintained using **virtualization** or containers
- Example: Multiple companies running applications on the same server in AWS.

Advantages and Disadvantages: Cloud providers use multi-tenancy because it:

- Improves **resource utilization-**i/o cpu overlap
- Reduces **cost for customers**
- But it also introduces **security risks.**

Security Risks in Multi-Tenancy

When multiple tenants share infrastructure, some risks may occur. Two common risks:

- **Data Leakage**
- **Noisy Neighbor Problem**

These risks happen because resources are shared among users.

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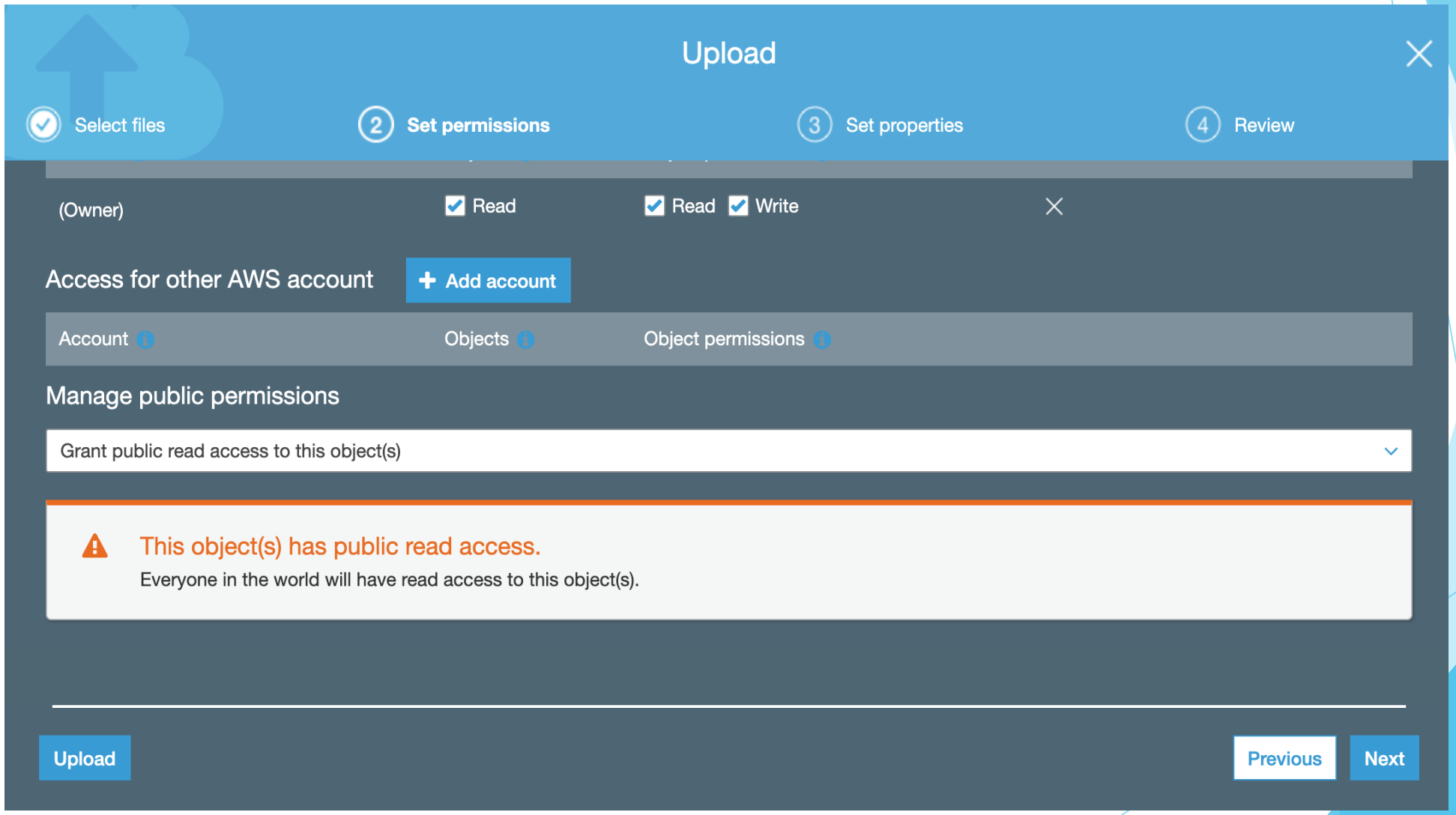
Data Leakage is the unauthorized transmission, exposure, or access of sensitive or confidential data to individuals, applications, or systems that are not permitted to view it. In a multi-tenant cloud environment, data leakage occurs when the data belonging to one tenant (customer or organization) becomes accessible to another tenant due to misconfiguration, vulnerabilities, or inadequate isolation mechanisms.

Example: If a storage bucket in a cloud platform (like AWS S3) is mistakenly configured with **public read access**, another tenant or user on the internet may be able to view or download sensitive files, resulting in data leakage.

In the AWS console, there is an option called **“Public access” / “Public read access”** that can expose files to everyone on the internet



Possible causes of Data Leakage

- **1. Misconfigured Storage:**
  Occurs when storage services (such as cloud buckets or databases) are configured with incorrect permissions, allowing unauthorized users to access sensitive data.
- **2. Weak Access Control:**
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  Happens when authentication and authorization mechanisms are poorly implemented, enabling users to access data or resources beyond their permitted privileges.
- **3. Software Vulnerabilities:**
  Security flaws or bugs in applications, operating systems, or cloud services that attackers can exploit to gain unauthorized access to sensitive information.
- **4. Improper Data Isolation:**
  In multi-tenant environments, inadequate separation between different tenants' data can allow one tenant to access or interfere with another tenant's data.

Noisy Neighbor Problem

The **Noisy Neighbor Problem** refers to a situation in a multi-tenant cloud environment where one tenant consumes an excessive amount of shared resources—such as CPU, memory, storage, or network bandwidth—thereby degrading the performance and availability of services for other tenants sharing the same infrastructure. This leads to performance degradation.

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- **Example:**

If multiple companies are running applications on the same cloud server and one company runs a very resource-intensive data processing job, it may consume most of the CPU and memory, causing the applications of other companies to slow down or respond poorly.

This happens because both tenants share the same physical server.

Possible causes of the Noisy Neighbor Problem

- **1. Excessive Resource Usage**
  One tenant runs heavy workloads (large computations, big data processing, or high traffic applications) that consume most of the shared CPU, memory, or network bandwidth.
- **2. Lack of Proper Resource Allocation**
  If the cloud provider does not properly limit or allocate resources for each tenant, one tenant may use more resources than intended.
- **3. Sudden Traffic Spikes**
  An application experiencing a sudden increase in user requests can consume shared infrastructure resources and slow down other tenants.
- **4. Inefficient Application Design**
  Poorly optimized applications may continuously consume high CPU, memory, or disk I/O, affecting the performance of other tenants.

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Mitigation of Multi-Tenancy Risks”

- “How Cloud Providers Reduce Multi-Tenancy Risks.”

• **Strong Virtualization Isolation**

Virtual machines and hypervisors isolate tenants so that one tenant cannot access another tenant’s resources.

• **Network Isolation:**Cloud providers isolate tenant networks using virtual networks so that one tenant’s traffic cannot reach another tenant’s resources. Private networks are created for each tenant

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• **Monitoring and Load Balancing**

Continuous monitoring detects abnormal resource usage and distributes workloads across servers to maintain performance.

• **Encryption and Access Control**

Data encryption and strict access control mechanisms protect sensitive data from unauthorized access.

Summary

- Multi-tenancy is an important feature of cloud computing but introduces some risks.
- Key risks:
- **Data Leakage** – unauthorized access to sensitive data
- **Noisy Neighbor** – performance issues due to resource sharing

Dr. Mustafa Ehsan

Cloud providers implement **security and isolation mechanisms** to minimize these risks.